

EXHIBIT 40

EXHIBIT 40

U.S. Patent No. 10,707,167 Infringement Chart

Title: Contacts to semiconductor substrate and methods of forming same

Inventors: Emre Alptekin, Nicolas L. Breil, Christian Lavoie, Ahmet S. Ozcan, Kathryn T. Schonenberg

Abstract: An aspect of the invention includes a method for forming a contact in a dielectric layer over a semiconductor substrate. The method may comprise: forming a contact opening in a dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.

Claim	Tower '167 Accused Products
[1pre] A semiconductor structure including a contact to a semiconductor substrate, the contact comprising:	To the extent the preamble is limiting, devices manufactured by Tower that rely on its SiGe and BiCMOS processes (including the Tower SBC18 process) (hereinafter, "Tower '167 Accused Products") comprise a semiconductor structure including a contact to a semiconductor substrate, as explained further in the limitations below. For example, the Toyota Blind Spot Monitor Sensor (#88162-0R031) includes a module manufactured by DENSO Corporation (identified as DNSRR004), which further includes a die manufactured by Tower (identified as M7DN25PS) that is, on information and belief, representative of devices manufactured by Tower that rely on its SBC18 process.¹

¹ GlobalFoundries reserves the right to identify to identify different or additional Tower '167 Accused Products and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances. Such supplemental information may include, but is not limited to, source code, data sheets, design specifications, deposition testimony, testing information, reference designs implementation and utilization information, and/or schematics.

EXHIBIT 40

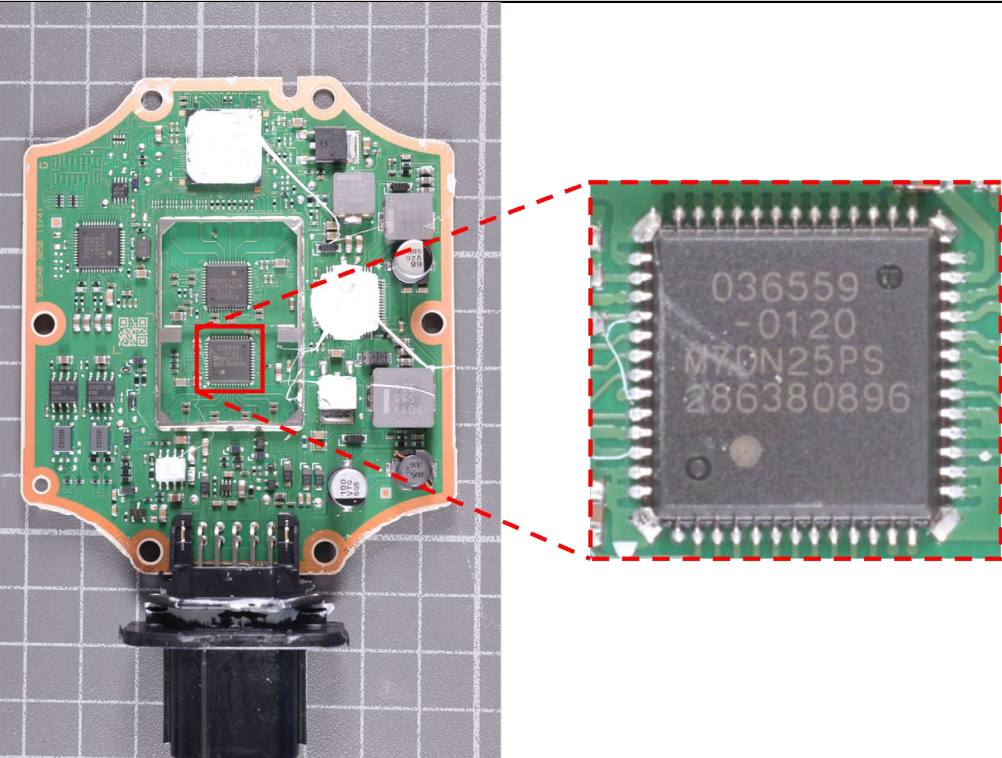
Claim	Tower '167 Accused Products
	
[1a] a contact opening in a dielectric layer over the semiconductor substrate;	The Tower '167 Accused Products comprise a contact opening in a dielectric layer over the semiconductor substrate. For example, a cross sectional analysis of the Tower '167 Accused Products reveals transistors with contact openings in a dielectric layer, as shown below.

EXHIBIT 40

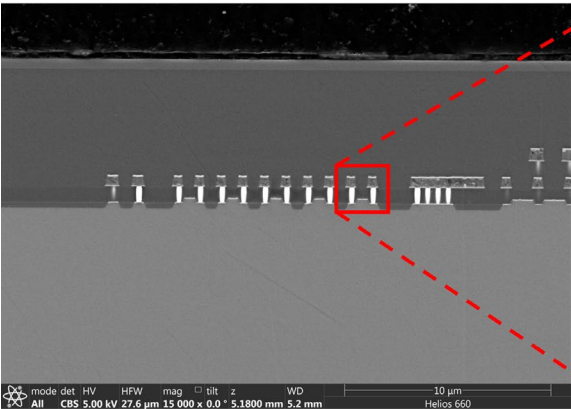
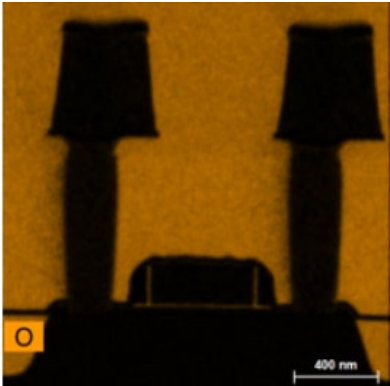
Claim	Tower '167 Accused Products
	   The first image is a scanning electron micrograph (SEM) showing a series of vertical structures on a substrate. A red box highlights one of these structures. The second image is a high-magnification SEM view of the structure highlighted in the first image, showing a contact opening. The third image is another SEM view of a similar structure, showing a contact opening. Both the second and third images include scale bars indicating 500 nm and 400 nm respectively.
[1b] a first liner layer substantially coating a sidewall of the contact opening;	The Tower '167 Accused Products comprise a first liner layer substantially coating a sidewall of the contact opening. For example, the reverse engineering analysis below shows a first liner layer on the sidewall of the contact opening.

EXHIBIT 40

Claim	Tower '167 Accused Products
	The images below resolve the elements overlaid in the material analysis above into separate elements.
[1c] a refractory metal layer substantially coating the first liner layer in the contact opening;	The Tower '167 Accused Products comprise a refractory metal layer substantially coating the first liner layer in the contact opening.

EXHIBIT 40

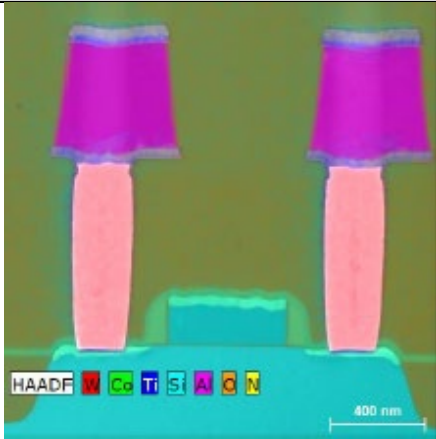
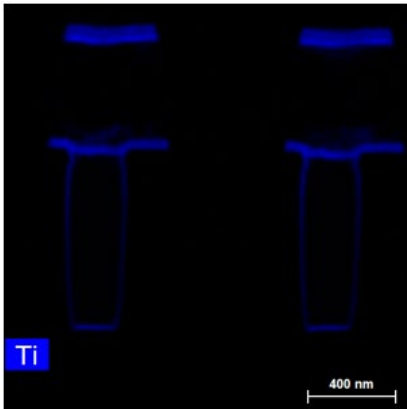
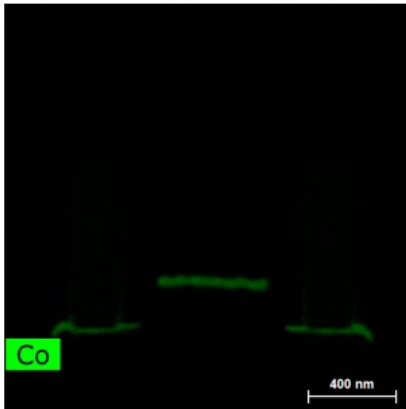
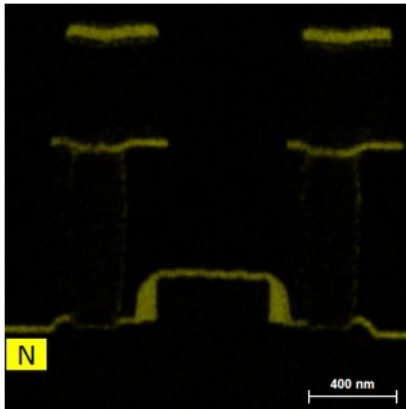
Claim	Tower '167 Accused Products
	 For example, the materials analysis below shows a refractory metal layer coating the first liner layer.   
[1d] a metal substantially filling the contact opening; and	The Tower '167 Accused Products comprise a metal substantially filling the contact opening.

EXHIBIT 40

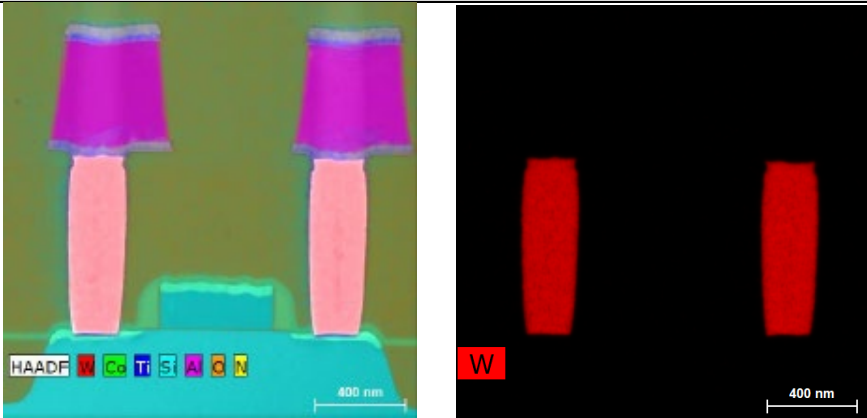
Claim	Tower '167 Accused Products
	
[1e] an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate;	The Tower '167 Accused Products comprise an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate. For example, the materials analysis charts below show the presence of an intermix region at an upper surface of the substrate in the contact opening. The green arrow in the TEM image below corresponds to the green arrow at the top of the ensuing materials analysis chart.

EXHIBIT 40

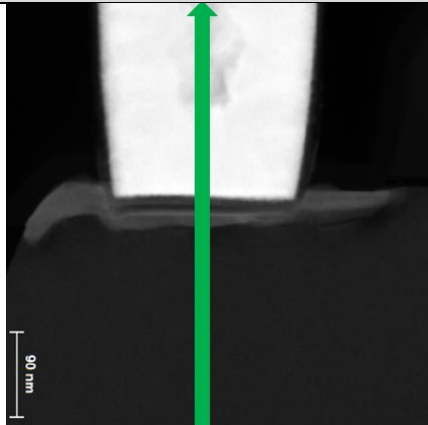
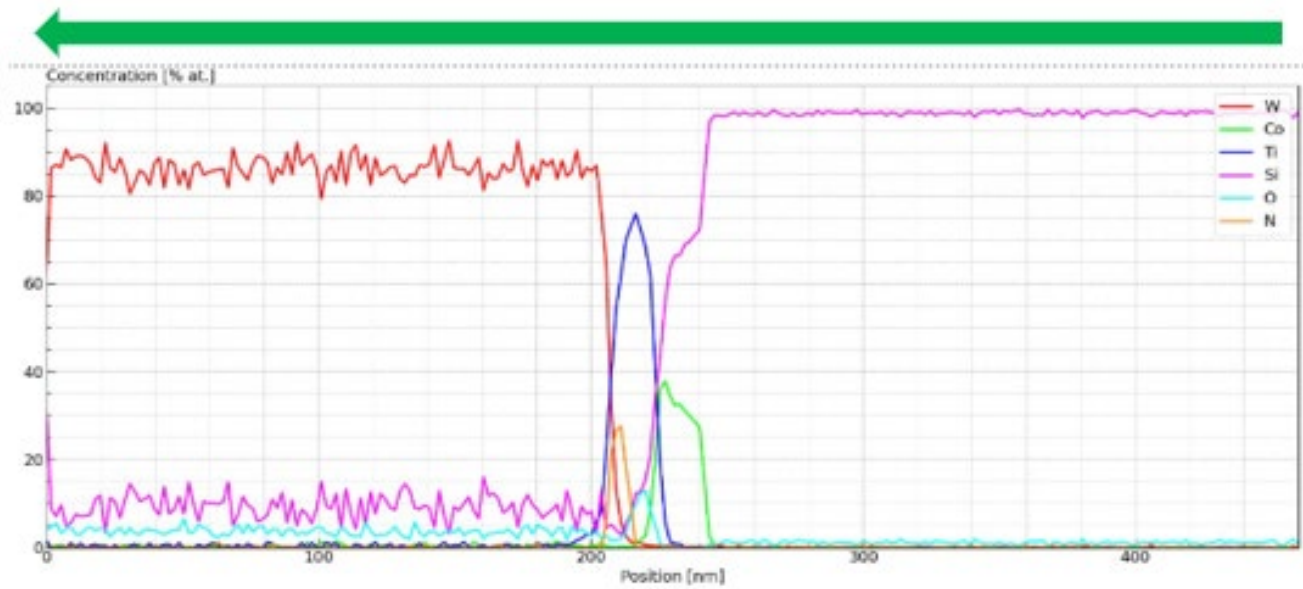
Claim	Tower '167 Accused Products
	 As shown in the material analysis chart below, the intermix region in the '167 Accused Products includes a material used in the first liner layer, the refractory metal layer, and the substrate. 

EXHIBIT 40

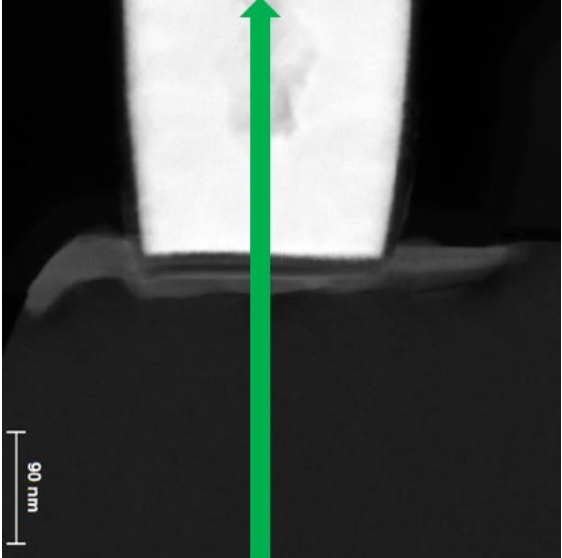
Claim	Tower '167 Accused Products
[1f] wherein the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.	The Tower '167 Accused Products comprise the refractory metal layer including a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal. For example, in the Tower '167 Accused Products, the refractory metal layer includes a first refractory metal material and a second refractory metal material, as shown in the material analysis chart below. The first refractory metal material substantially separates the first liner layer and the second refractory metal material. Likewise, the second refractory metal material substantially separates the first refractory metal material and the metal. 

EXHIBIT 40

